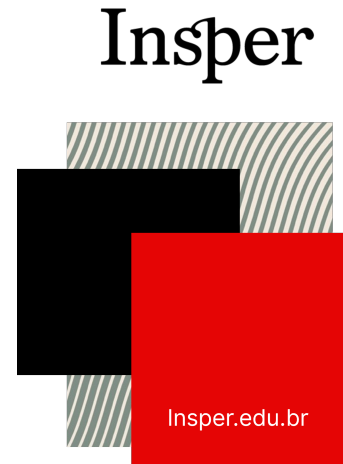
Economics of Organizations and Incentives

MPE – 4th term, 2026

Lecture 4: Credible threats and subgame perfection



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Today

1. Case: Boris's Brexit dilemma.
2. Nash equilibrium is not enough: non-credible threats.
3. Subgames and subgame-perfect equilibrium.
4. Repeated games: definition.

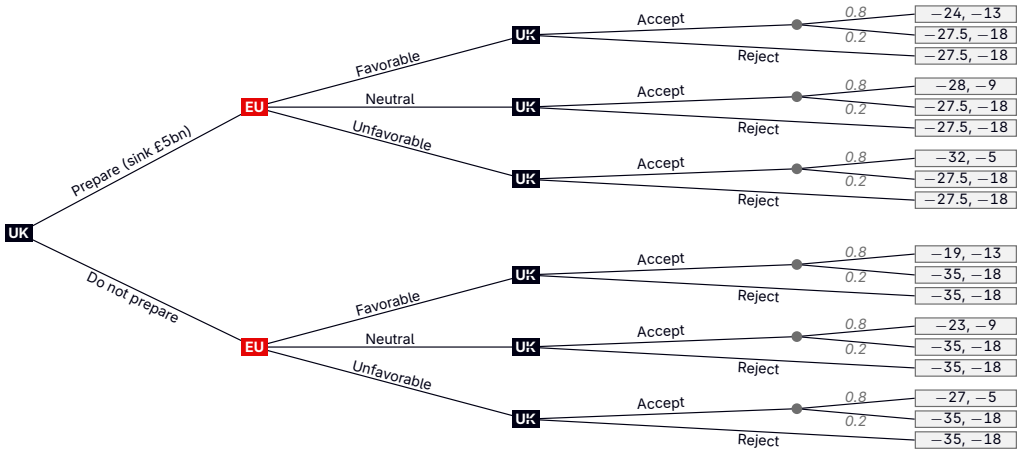
Case: Boris's Brexit dilemma

Boris's Brexit dilemma: the data

Cost (£bn; payoff $= -$cost)	UK	EU
Hard Brexit, UK not prepared	35	18
Hard Brexit, UK prepared	27.5 $(= 35 - 12.5 + 5)$	18
Favorable deal	19	13
Neutral deal	23	9
Unfavorable deal	27	5

- Preparation costs £5bn (all or nothing), mitigates £12.5bn of a hard Brexit, and is *sunk*: if there is a deal, add £5bn to the UK's cost.
- Parliament approves a deal brought to it with probability 0.8. If Parliament rejects it, or if the UK rejects the EU's offer outright, the outcome is a hard Brexit.

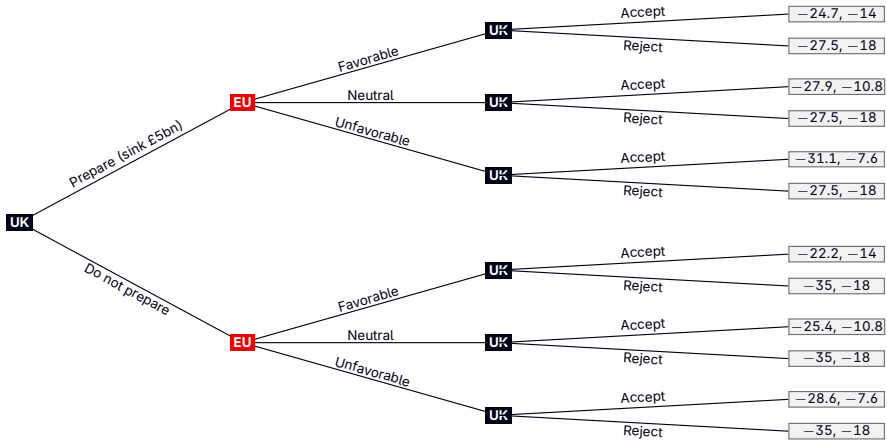
The Brexit game in extensive form



Source: adapted from Exhibit TN2, *Boris's Brexit Dilemma (A), (B), and (C)*, Teaching Note UV9622, Darden Business Publishing, 2023.

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Replacing Nature by expected payoffs



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The Brexit game in strategic form

	Player 2: EU								
Player 1: UK	**UU**	**UN**	**UF**	**NU**	**NN**	**NF**	**FU**	**FN**	**FF**
PU	−31.1, −7.6	−31.1, −7.6	−31.1, −7.6	−27.9, −10.8	−27.9, −10.8	−27.9, −10.8	−24.7, −14	−24.7, −14	−24.7, −14
PN	−27.5, −18	−27.5, −18	−27.5, −18	−27.9, −10.8	−27.9, −10.8	−27.9, −10.8	−24.7, −14	−24.7, −14	−24.7, −14
PF	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−24.7, −14	−24.7, −14	−24.7, −14
P0	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18
DU	−28.6, −7.6	−25.4, −10.8	−22.2, −14	−28.6, −7.6	−25.4, −10.8	−22.2, −14	−28.6, −7.6	−25.4, −10.8	−22.2, −14
DN	−35, −18	−25.4, −10.8	−22.2, −14	−35, −18	−25.4, −10.8	−22.2, −14	−35, −18	−25.4, −10.8	−22.2, −14
DF	−35, −18	−35, −18	−22.2, −14	−35, −18	−35, −18	−22.2, −14	−35, −18	−35, −18	−22.2, −14
D0	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18

UK: P/D = prepares / does not prepare; then accepts any deal at least as good as U (unfavorable), N (neutral), F (favorable), or 0 = rejects everything (hard Brexit).
EU: first letter = offer if the UK prepared; second letter = offer if the UK did not prepare.

The Brexit game in strategic form

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Player 1: UK	**UU**	**UN**	**UF**	**NU**	**NN**	**NF**	**FU**	**FN**	**FF**
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PN	−27.5, −18	−27.5, −18	−27.5, −18	−27.9, −10.8	−27.9, −10.8	−27.9, −10.8	−24.7, −14	−24.7, −14	−24.7, −14
PF	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−24.7, −14	−24.7, −14	−24.7, −14
P0	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18
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UK: P/D = prepares / does not prepare; then accepts any deal at least as good as U (unfavorable), N (neutral), F (favorable), or 0 = rejects everything (hard Brexit).
EU: first letter = offer if the UK prepared; second letter = offer if the UK did not prepare.

Find the Nash equilibria.

Source: adapted from Exhibit TN2, *Boris's Brexit Dilemma (A), (B), and (C)*, Teaching Note UV9622, Darden Business Publishing, 2023.

Too many Nash equilibria

	Player 2: EU								
Player 1: UK	**UU**	**UN**	**UF**	**NU**	**NN**	**NF**	**FU**	**FN**	**FF**
PU	−31.1, −7.6	−31.1, −7.6	−31.1, −7.6	−27.9, −10.8	−27.9, −10.8	−27.9, −10.8	−24.7, −14	−24.7, −14	−24.7, −14
PN	−27.5, −18	−27.5, −18	−27.5, −18	−27.9, −10.8	−27.9, −10.8	−27.9, −10.8	−24.7, −14	−24.7, −14	−24.7, −14
PF	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	**−24.7, −14**	**−24.7, −14**	−24.7, −14
P0	**−27.5, −18**	−27.5, −18	−27.5, −18	**−27.5, −18**	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18
DU	−28.6, −7.6	−25.4, −10.8	−22.2, −14	−28.6, −7.6	−25.4, −10.8	−22.2, −14	−28.6, −7.6	−25.4, −10.8	−22.2, −14
DN	−35, −18	**−25.4, −10.8**	−22.2, −14	−35, −18	**−25.4, −10.8**	−22.2, −14	−35, −18	−25.4, −10.8	−22.2, −14
DF	−35, −18	−35, −18	**−22.2, −14**	−35, −18	−35, −18	**−22.2, −14**	−35, −18	−35, −18	**−22.2, −14**
D0	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18	−35, −18

There are 9 pure-strategy Nash equilibria (shaded).

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Player 1: UK	**UU**	**UN**	**UF**	**NU**	**NN**	**NF**	**FU**	**FN**	**FF**
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PN	−27.5, −18	−27.5, −18	−27.5, −18	−27.9, −10.8	−27.9, −10.8	−27.9, −10.8	−24.7, −14	−24.7, −14	−24.7, −14
PF	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−24.7, −14	−24.7, −14	−24.7, −14
P0	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18
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DN	−35, −18	−25.4, −10.8	−22.2, −14	−35, −18	−25.4, −10.8	−22.2, −14	−35, −18	−25.4, −10.8	−22.2, −14
DF	−35, −18	−35, −18	−22.2, −14	−35, −18	−35, −18	−22.2, −14	−35, −18	−35, −18	**−22.2, −14**
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There are 9 pure-strategy Nash equilibria (shaded).

- (DF, FF) (in bold): the UK does not prepare and “accepts only a favorable deal”; the EU offers a favorable deal. Is this threat **credible**?

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Player 1: UK	**UU**	**UN**	**UF**	**NU**	**NN**	**NF**	**FU**	**FN**	**FF**
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P0	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18	−27.5, −18
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- (DF, FF) (in bold): the UK does not prepare and "accepts only a favorable deal"; the EU offers a favorable deal. Is this threat **credible**?
- The threat is tested in column FU (the EU offers U if the UK did not prepare): the UK gets -35 with DF (rejecting) but -28.6 with DU (accepting), in red.

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- (DF, FF) (in bold): the UK does not prepare and "accepts only a favorable deal"; the EU offers a favorable deal. Is this threat **credible**?
- The threat is tested in column FU (the EU offers U if the UK did not prepare): the UK gets -35 with DF (rejecting) but -28.6 with DU (accepting), in red.
- In column FF, however, DF and DU give the same payoff: U is never offered, so the strategic form cannot tell whether the threat is credible. **We need a refinement.**

Source: adapted from Exhibit TN2, *Boris's Brexit Dilemma (A), (B), and (C)*, Teaching Note UV9622, Darden Business Publishing, 2023.

Backward induction (1): the UK's last move

Compare accepting (expected payoff) with rejecting (hard Brexit):

	UK prepared		UK not prepared	
Offer	Accept	Reject	Accept	Reject
Favorable	$-$**24.7**	-27.5	$-$**22.2**	-35
Neutral	-27.9	$-$**27.5**	$-$**25.4**	-35
Unfavorable	-31.1	$-$**27.5**	$-$**28.6**	-35

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Favorable	$-$**24.7**	-27.5	$-$**22.2**	-35
Neutral	-27.9	$-$**27.5**	$-$**25.4**	-35
Unfavorable	-31.1	$-$**27.5**	$-$**28.6**	-35

- Not prepared: the UK accepts *any* deal.
- Prepared: the UK accepts *only* the favorable deal. Preparation makes the threat to reject credible.

Backward induction (2): earlier moves

EU, anticipating the UK's responses:

- UK prepared: favorable (-14) $\succ$ neutral or unfavorable, which are rejected (-18).

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UK, at the first node: prepare gives -24.7; not prepare gives -28.6.
Prepare.

Backward-induction solution

UK: prepare; if prepared, accept only the favorable deal; if not prepared, accept any deal.
EU: favorable deal if the UK prepared; unfavorable deal otherwise (FU).
Outcome: the UK prepares and receives a favorable deal: payoffs (-24.7, -14).

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Lessons from the case

- **Sunk investments can be strategic.** The £5bn is “wasted” if there is a deal, but it is what makes the deal good: it improves the UK’s outside option and makes rejection credible.
- **Observability matters.** The investment works only because the EU sees it. What if preparation were secret?

Lessons from the case

- **Sunk investments can be strategic.** The £5bn is "wasted" if there is a deal, but it is what makes the deal good: it improves the UK's outside option and makes rejection credible.
- **Observability matters.** The investment works only because the EU sees it. What if preparation were secret?
- **Talk is cheap.** Announcing "I will reject a bad deal" without preparing is not credible: the EU knows the UK would accept.

Subgames and subgame-perfect equilibrium

From the case to a definition

- In the Brexit game, several Nash equilibria rely on threats that would not be carried out if the moment came (“I accept only a favorable deal” without preparing).
- Nash equilibrium only requires optimal behavior *on* the equilibrium path. Off the path, a player may plan to do something that is not in her interest.
- Backward induction rules these threats out: each player must play optimally at every node, including those that are never reached.

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To generalize this idea to games with imperfect information, we need the notion of a **subgame**.

Subgames

Definition (subgame)

A **proper subgame** G of an extensive-form game Γ consists of a single node and all its successors in Γ, with the property that if $x \in G$ and $x' \in h(x)$, then $x' \in G$. The subgame G is itself a game tree, with information sets and payoffs inherited from Γ.

Subgames

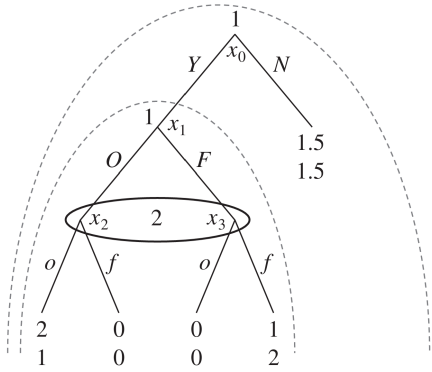
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In words:

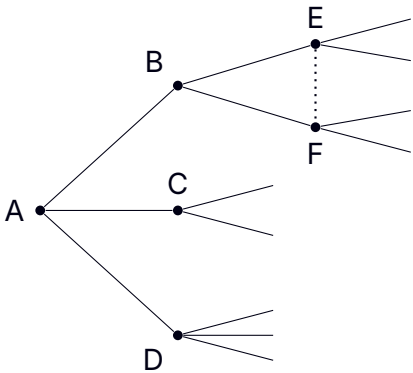
- a subgame starts at a single node, whose information set contains only that node;
- it contains everything that can happen from that node on;
- it does not "cut" any information set.

Subgames



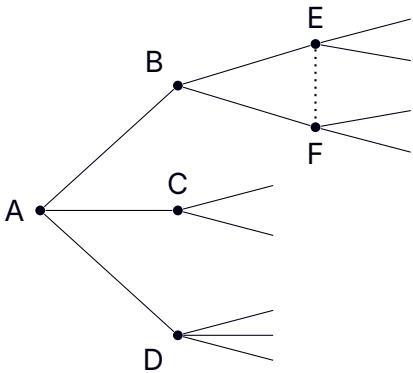
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Example 1: how many subgames?



How many subgames does this game have?

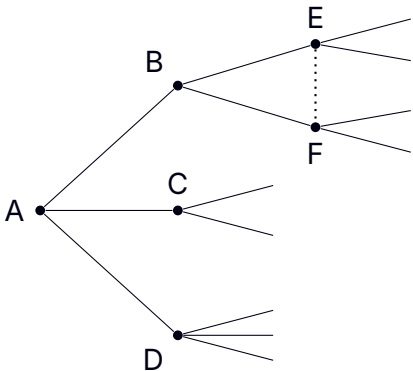
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How many subgames does this game have?

Four: the game itself (from A) and the subgames that start at B, C and D.

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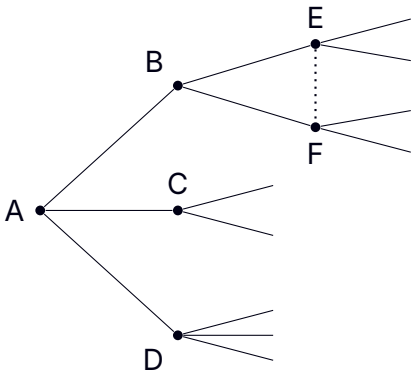


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E and F do *not* start subgames: they belong to the same information set, so a subgame starting at E would cut that information set.

Example 1: how many subgames?



How many subgames does this game have?

Four: the game itself (from A) and the subgames that start at B, C and D.

E and F do *not* start subgames: they belong to the same information set, so a subgame starting at E would cut that information set.

B *does* start a subgame: the information set $\{E, F\}$ lies entirely inside it.

Subgame-perfect equilibrium

Definition (subgame-perfect equilibrium)

Let Γ be an n-player extensive-form game. A behavioral strategy profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a **subgame-perfect (Nash) equilibrium** if, for every proper subgame G of Γ, the restriction of σ^* to G is a Nash equilibrium in G.

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- Every subgame-perfect equilibrium is a Nash equilibrium (the whole game is a subgame).
- Not every Nash equilibrium is subgame perfect: the concept **refines** Nash equilibrium by ruling out non-credible threats.

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- Every subgame-perfect equilibrium is a Nash equilibrium (the whole game is a subgame).
- Not every Nash equilibrium is subgame perfect: the concept **refines** Nash equilibrium by ruling out non-credible threats.

Fact: in finite games of perfect information, the subgame-perfect equilibria are exactly the backward-induction solutions.

Back to the Brexit game

- How many subgames does the Brexit game have?

Back to the Brexit game

- How many subgames does the Brexit game have?
- **9:** the whole game, one at each EU node (2), and one at each UK node after an offer (6).

Back to the Brexit game

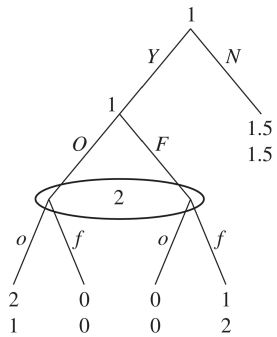
- How many subgames does the Brexit game have?
- **9:** the whole game, one at each EU node (2), and one at each UK node after an offer (6).
- Of the 9 Nash equilibria of the strategic form, only the backward-induction solution is a Nash equilibrium in every subgame.

Back to the Brexit game

- How many subgames does the Brexit game have?
- **9:** the whole game, one at each EU node (2), and one at each UK node after an offer (6).
- Of the 9 Nash equilibria of the strategic form, only the backward-induction solution is a Nash equilibrium in every subgame.
- For example, (DF, FF) fails in the subgame that starts after “no preparation, unfavorable offer”: rejecting is not a best response there.

Example 2

Find all Nash equilibria and all subgame-perfect equilibria of the *voluntary* battle of the sexes:



Finitely repeated games

Repeated games

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- workers collaborating daily in a team; a firm and its suppliers.

Question: can the shadow of the future sustain behavior that is not an equilibrium of the one-shot game, such as cooperation in the prisoner's dilemma?

Finitely repeated games

Definition

Given a stage game G, the **finitely repeated game** $G(T, \delta)$ is the game in which G is played in T consecutive periods, with common discount factor $\delta \in (0, 1]$. Player i's payoff is

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- At the start of each period, players observe the actions chosen in all previous periods (the **history**).
- A strategy says what to play in each period *after each possible history*.
- Each period starts a new subgame.

Wrap-up and next lecture

Today

- Boris's Brexit dilemma: credible threats, sunk investments and outside options.
- Non-credible threats in Nash equilibrium; subgames and subgame-perfect equilibrium.
- Repeated games: definition and payoffs.

Next lecture

- Finitely repeated games: unraveling, carrots and sticks.
- Infinitely repeated games, the one-stage deviation principle and the folk theorem.
- Bargaining.